

Cognitive Architectures for Robots inspired by Quantum Computing (Tentative Description)

by

Yeshwanth guru krishnakumar



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**Università
di Genova**

University of Genoa

ABSTRACT:

This project is to design and develop a novel version of a framework(quantum robot),to design Cognitive Architecture for robots inspired by quantum Computing. This project is to implement on Tiago robot with a simple Handover task(1.0). In this scene, we have a Tiago robot, a human and an object. As the robot pick and handover (object) to the person.

INTRODUCTION:

This whole project scene is developed in the Gazebo simulation environment, which was simulated based on the nodes developed using **ROS MELODIC(ubuntu 18.04)**. The main aim of the project is to define a Software Architecture(ROS) to control the robot in gazebo simulation. In particular we should plan robot movement using the Moveit Package. And let the Tiago , pick the object and handover to the person. where we have the “Decision making” Quantum system node which gets the multi sensory inputs from the defined inputs from the simulation environment when the inputs are measure it triggers the picking system.

TENTATIVE ARCHITECTURE OF THE SYSTEM:

In the following UML diagram is shown the whole Cognitive architecture.

Our system to control the Tiago robot behavior consists of Five nodes. The node `go_to_home` is not shown in the UML diagram since it is not really used in the simulation but only if we have the robot in a random position and we want to move it in the home position. On the Down side of the diagram we have the Gazebo simulation that communicates with the ROS nodes that are on the up side. The simulation sends information to ROS using the `State_machine`. The `state_machine` sends the information about the position of the object that to be handover, the hand of the human and the joint configuration. Those information are sent to the task planner whose object is to decide what each arm should do. If the task planner finds at least one task for one arm, a task is sent as a request to the server `TIAGO`. The server tries and find a feasible trajectory using `MoveIt`. The `TIAGO` node returns true if a trajectory as soon as a feasible trajectory has been found, it sends false if no trajectory is acceptable. In order for the task planner to know if the task has been completed and to decide what to do next we need the `Tiago at the Home` node.

The robot nodes receives from Gazebo the position of the joints and checks whether they are the same as the home configuration, only in that case it returns true to the task planner that can move on and find the new trajectory. In the next page, we have the component UML and the sequential UML.

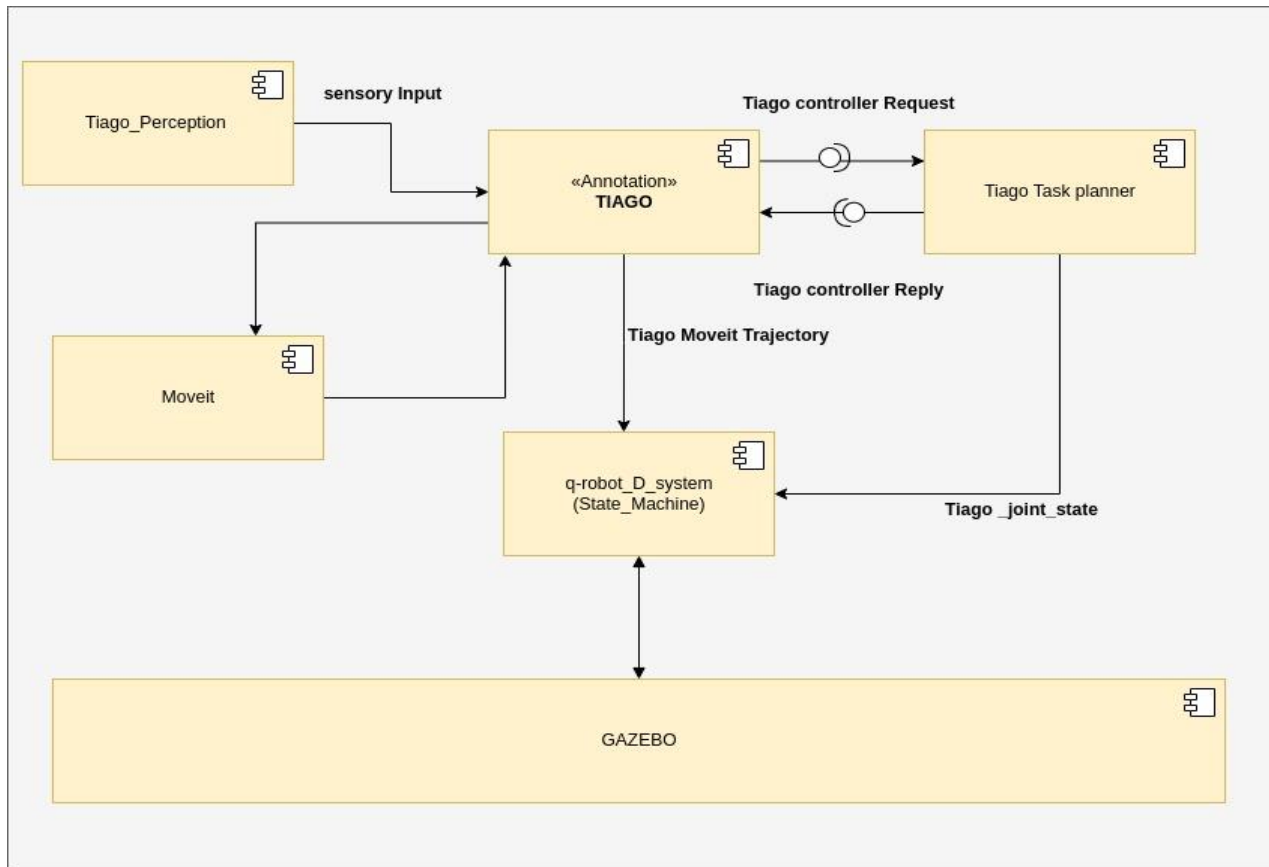


Fig:1:TENTATIVE COMPONENT DIAGRAM

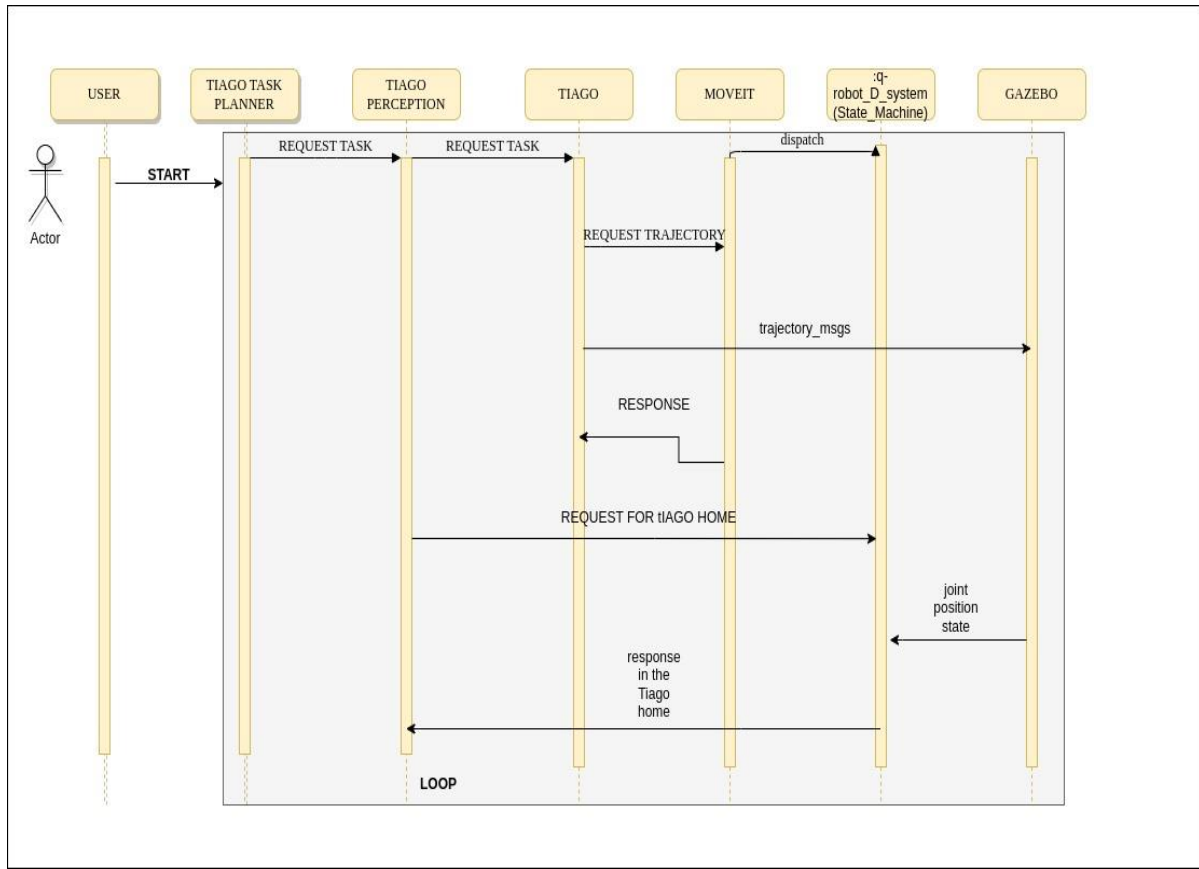


Fig:2:TENTATIVE SEQUENCE DIAGRAM

TENTATIVE DESCRIPTION OF THE ROBOT ARCHITECTURE AND ITS WORKING:

Our architecture is a classic hierarchical arch. Initially we wanted to implement a hybrid reactive-deliberate architecture which include the q-robot decision system with Multisensory input in which our main aim to Trigger the picking system ,which defined from the simulation working environment. As the object moves fastly in between the robot hand and the human hand, when the object has "enough time" on the right (based on how we set the temporal windows), the picking system is triggered as long as the quantum system continues to decide "go". In this way, should be pretty simple to associate $|0\rangle$ with the robot and $|1\rangle$ with the human hand, and activate the picking mechanism every time we measure $|1\rangle$ and keeping it activated as long as we read $|1\rangle$.

TIAGO_TASK_PLANNER & TIAGO_MAIN: The series of task_planner and Tiago_main is a configuration very similar to a computational pattern, which would plan the working task for robot.

TIAGO_PERCEPTION: To Analyze the multi-sensory Integration based upon the simulation Environment configuration which we defined in this node through the cognitive specification. It can be done based on some experiments, such as

Case 1: In order to make it easy to check this orientation, Markers in rviz were used. This way it is possible to check it visually. Therefore, for performing this experiment, the perception module should be run. The perception module detects the objects and estimates their position, then publishes in a ROS topic the position and orientation using a MarkerArray message. Finally, it is possible to compare the orientation of each labeled can, shown in the capture done by perception module, with the orientation displayed with the **rviz markers**.

Case 2: With this experiment we want to compare the positions given by the perception module with the positions given by a typical and very used pose estimation algorithm, as the **Aruco markers detector**.

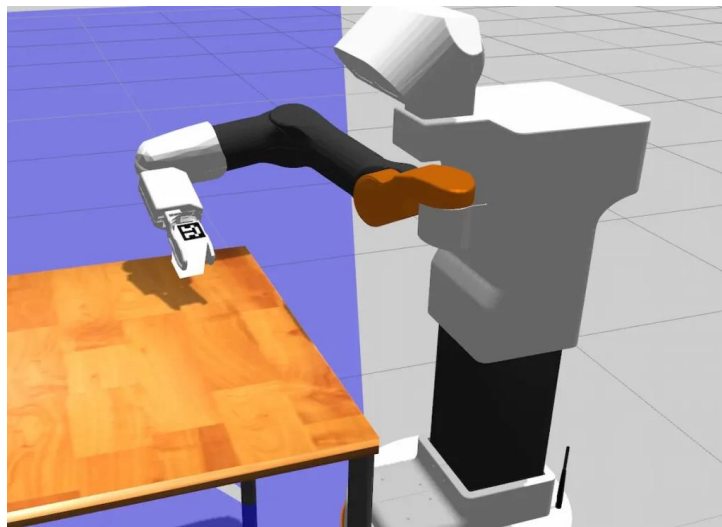


Fig 3: Aruco marker usage (expected)

Case 3: On the other hand, the position of the cans of soda obtained from the perception algorithm, are compared with the position of the end effector. This way, it is possible to check if the real positions of the end effector and the cans correspond with the positions shown in rviz. This is useful to check if the path planner will be able to reach to can's position before executing the movement, just moving by hand the end effector to the can's position.

MOVEIT: Then Moveit! the package is used in order to plan a pick trajectory to grasp the object and execute the task.

Q-ROBOT-D-SYSTE~~M~~(state_machine): It takes the data from Gazebo and returns them in a format compatible for our ROS Nodes, it also involved in as the pub-sub communication pattern as Publisher where the Quantum system works to execute in the simulation.

GAZEBO:Final execution of the simulation .

EXAMPLE FOR EXPECTED SIMULATION SCENE :

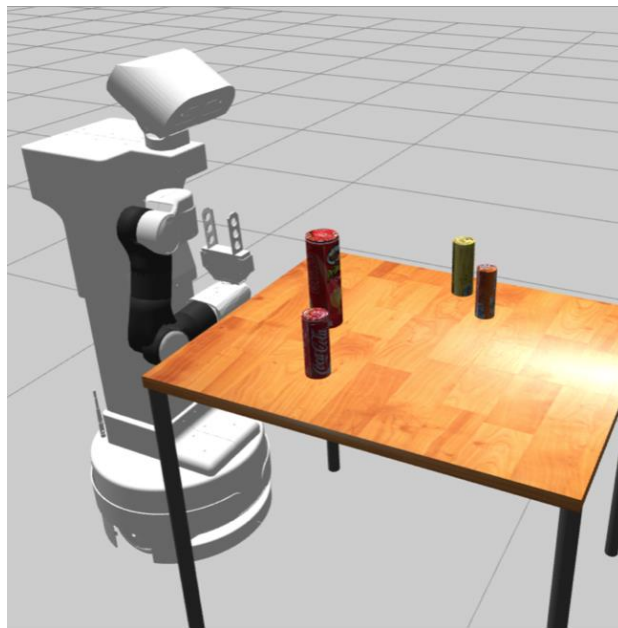


Fig 4:Simulation in Gazebo

4. INSTALLATION:

As the tiago robot supports the ROS melodic (ubuntu18.04) needed to make the setup where Ros melodic can be installed from the [ROS MELODIC](#) and the ubuntu from the [UBUNTU 18.04](#).It should be followed up with the moveit package from [ROS MELODIC MOVEIT PACKAGE](#).

Tiago robot package should be downloaded from the [Tiago robot package](#).And the q-robot package should be downloaded based upon the description in the following link [Q-robot](#).